

6G Weekly Survey — Jun 29 - Jul 05, 2026

15 items across 5 topics, ranked by importance.

AI-native air interface

Toward AI-Native 6G Air Interface: A 3GPP Perspective on Protocol Framework

Xingqin Lin · arXiv cs.NI · 2026-06-25

★★★★★ This directly addresses the critical standardization question for AI-native 6G and comes from a credible 3GPP perspective, providing actionable protocol-level guidance that will likely influence actual specification work. This paper argues that making the 6G air interface truly AI-native requires standardizing the protocol framework for coordinating AI-enabled radio functions across vendors, rather than prescribing specific model architectures or training methods. The author presents a 3GPP-oriented perspective on how to configure, validate, activate, monitor, and safely revert AI functions while preserving implementation freedom. A neural receiver case study illustrates reference signal adaptation as an example of this architectural approach.

Contribution: Proposes a concrete 3GPP-aligned protocol framework for AI-native air interface that focuses on interoperability mechanisms (configuration, validation, monitoring) rather than mandating specific AI/ML implementations.

Relevance to your work: Directly relevant to your AI-native air interface work—this provides the standards-level framework for how neural receivers and other learned PHY components you research would actually be deployed and coordinated in multi-vendor 6G networks.

Channel modeling / GAN-based

Enormous Fluid Antenna Systems (E-FAS) for Wireless Sensing: Channel Modeling and Conditional Estimation Limits

Farshad Rostami Ghadi, Kai-Kit Wong, Jose D. Vega-Sanchez, Kin-Fai Tong, et al. · arXiv cs.IT · 2026-06-22

★★★★■ Introduces a new ISAC architecture with rigorous channel modeling and fundamental performance bounds, addressing an emerging reconfigurable surface paradigm with clear differentiation from RIS and conventional arrays. This paper develops an analytical framework for Enormous Fluid Antenna Systems (E-FAS) applied to integrated sensing and communications, modeling the complete bidirectional sensing channel including surface-wave routing, distributed reradiation, target scattering, and echo propagation. The authors derive Fisher information and Cramér-Rao bounds for angular target estimation, revealing a fundamental trade-off between surface-wave routing gain and sensing diversity that differs from conventional array or RIS-based ISAC. Numerical results show E-FAS can achieve substantial angular sensing gains over conventional architectures under equal transmit power.

Contribution: First analytical framework establishing that E-FAS creates a fundamentally different sensing regime where maximizing coherent routing gain does not maximize sensing performance, exposing a novel routing-diversity trade-off in programmable propagation environments.

Relevance to your work: Directly relevant to your ISAC research interests and complements your channel modeling work; the parametric observation model and fundamental limits analysis could inform future learning-based sensing approaches in programmable electromagnetic environments.

AI-Empowered UAV-Assisted Backscatter Localization and ISAC for Zero-Energy IoT: A Comprehensive Survey

Ruhul Amin Khalil · arXiv eess.SP · 2026-06-22

★★★ ■ Solid synthesis of an emerging niche (UAV-backscatter-ISAC) relevant to 6G IoT verticals, but primarily consolidates existing work rather than introducing novel methods or datasets for beyond-5G physical layer research.

This survey examines UAV-assisted backscatter communication for zero-energy IoT, where UAVs serve as mobile carrier emitters and receivers to overcome weak reflections and limited coverage in passive RF systems. The work unifies network architectures, backscatter modes, localization methods, and ISAC frameworks through a PRISMA-informed taxonomy, emphasizing AI-driven optimization for resource allocation and positioning. The authors identify open challenges in realistic channel modeling, energy-neutral operation, and integration with reconfigurable intelligent surfaces and 6G systems.

Contribution: This is a comprehensive taxonomy and structured survey of UAV-backscatter-ISAC systems for zero-energy IoT, covering multiple dimensions (network roles, AI techniques, sensing functions) with quantitative trend analysis and tutorial examples.

Relevance to your work: Relevant to your ISAC and vertical-applications interests; the discussion of channel modeling gaps and RIS/6G integration aligns with your work on AI-native air interfaces, though the focus is IoT/localization rather than core PHY design.

6G Demos & Trials

An Experimental Assessment of the Spatial and Frequency Selectivity of Reconfigurable Intelligent Surfaces

Cyrille Morin, Leonardo S Cardoso, Maxime Guillaud, Ahmad Shokair, et al. · arXiv cs.NI · 2026-06-25

★★★★■ Addresses a critical practical deployment challenge for RIS-aided 6G systems by revealing non-trivial interference coupling that could constrain multi-user RIS architectures and require coordination protocols.

This paper experimentally evaluates how a reconfigurable intelligent surface (RIS) optimized for one communication link affects nearby secondary links, using a Greenerwave RIS and CorteXlab testbed in FR1 bands. The authors test three scenarios with varying spatial and frequency separation between primary and secondary links. Results show significant impact on secondary link received power and channel phase even outside the RIS nominal frequency range, with carrier frequency separation providing negligible isolation.

Contribution: First experimental demonstration that RIS interference extends beyond nominal frequency ranges and spatial beam patterns, indicating coexistence challenges in multi-user RIS deployments.

Relevance to your work: Directly relevant to your RIS and ISAC work—experimental evidence of spatial/spectral coupling informs RIS beamforming design, multi-user scheduling, and interference management in AI-native air interfaces where RIS will be a key enabling technology.

Which Nash Equilibrium? Solver-Dependent Selection on Zero-Sum Nash Polytopes

Luis Leal · arXiv cs.LG · 2026-06-26

★★■ ■ ■ While mathematically rigorous for game theory, this work addresses solver behavior in abstract two-player games rather than wireless resource allocation or communication system design where multiple equilibria arise.

This paper investigates whether different Nash equilibrium solvers (CFR, CFR+, R-NaD, magnetic mirror descent) converge to different equilibria when multiple Nash equilibria exist in two-player zero-sum games. Using six analytically tractable games including Kuhn poker and a 180-game ensemble, the authors find that regularized last-iterate methods (R-NaD, magnetic mirror descent) consistently select the maximum-entropy equilibrium while regret-averaging methods (CFR variants) drift to lower-entropy boundary solutions. The selected equilibrium has measurable downstream consequences against suboptimal opponents, particularly in sequential games with hidden information.

Contribution: First systematic demonstration that Nash solver choice deterministically (not randomly) affects which equilibrium is selected from a Nash polytope, with regularized methods provably selecting maximum-entropy equilibria and CFR-family methods drifting to polytope boundaries.

Relevance to your work: Tangential—could inform multi-agent learning for beam management or interference games in beyond-5G if those scenarios exhibit Nash polytopes, but the testbed (poker, matrix games) and focus on solver comparison have no direct connection to AI-native air interfaces or physical-layer learning.

AB-Sync: Attention-Based Slot-Level Clock Synchronization Method for UWB-TDOA Localization Networks

Tianyi Lyu, Kefei Tian, Kangqiao Qin, Qingwen Liu, et al. · arXiv cs.NI · 2026-06-26

★★■ ■ ■ ■ Competent engineering solution for a narrow UWB-TDOA indoor positioning problem with demonstrated testbed improvements, but limited scope beyond specialized localization applications and no direct 6G air interface implications.

The paper addresses clock synchronization in UWB time-difference-of-arrival (TDOA) localization networks, where existing methods synchronize at coarse interval levels while TDMA systems localize tags in discrete short slots, creating granularity mismatch. AB-Sync uses attention mechanisms to model slot-specific clock-speed ratios from neighboring clock-fluctuation observations, enabling tag-slot-level timestamp mapping without additional synchronization messages. Real testbed results show 9.4% reduction in TDOA ranging standard deviation and 18.6% improvement in static localization accuracy versus baseline methods.

Contribution: First attention-based method to perform slot-level clock synchronization in UWB-TDOA systems, resolving the granularity mismatch between synchronization intervals and TDMA slot structure without additional overhead.

Relevance to your work: Tangential—focuses on indoor UWB positioning synchronization rather than 6G air interface design, AI-native PHY, or cellular sensing/communication integration.

RECAST: Model Reconstruction via Counterfactual-Aware Wasserstein Geometry under Limited Data

Xuan Zhao, Lena Krieger, Zhuo Cao, Arya Bangun, et al. · arXiv cs.LG · 2026-06-26

★★■ ■ ■ ■ While technically sound for ML auditing and explainability, this work focuses on generic black-box model reconstruction rather than wireless-specific problems, making it peripheral to 6G physical layer research despite appearing in '6G Demos & Trials' topic matches.

This paper addresses black-box model reconstruction using counterfactual explanations under limited data access, proposing RECAST which uses Wasserstein barycentric prototypes to build surrogate models that avoid decision boundary shifts and overfitting. The method incorporates counterfactuals as informative samples for both classes and enables fairness auditing without requiring online query access. Experiments demonstrate high fidelity and robustness in low-sample, noisy-access scenarios.

Contribution: RECAST enables faithful surrogate model reconstruction from black-box systems using counterfactuals and Wasserstein geometry without online access, addressing decision boundary shifts in low-data regimes while supporting fairness diagnostics.

Relevance to your work: Tangential—the methodology could theoretically apply to auditing learned PHY components or neural receivers as black boxes, but the paper contains no wireless systems context or demonstration.

A Beamforming Microwave Interferometric Radiometer for High-resolution Passive Imaging: Concept, Modeling, and Preliminary Demonstration

Ziyang Zhang, Hao Liu, Donghao Han, Te Wang, et al. · arXiv eess.SP · 2026-06-26

★★■ ■ ■ ■ Competent engineering solution for Earth observation radiometry but focused on passive sensing rather than communication or ISAC, with narrow scope limited to proof-of-concept for meteorological applications.

This paper proposes a beamforming microwave interferometric radiometer (BF-MIR) that uses beamforming-capable antennas as interferometric elements in sparse arrays to achieve high-resolution passive microwave imaging for Earth observation. The approach enlarges spatial-frequency sampling intervals to reduce array complexity while using narrow-beam spatial filtering to suppress aliasing from undersampling, with dynamic beam steering to compensate for limited instantaneous coverage. A three-element prototype demonstrates feasibility of the architecture for future spaceborne applications.

Contribution: The novel integration of beamforming antennas into interferometric radiometer arrays enables sparse array designs with reduced correlation complexity while maintaining high resolution through dynamic beam steering and aliasing suppression strategies.

Relevance to your work: Tangential—while beamforming and sparse arrays connect to massive MIMO and the prototype demonstrates beam management concepts, the passive radiometry focus for weather sensing does not directly advance AI-native air interfaces or integrated sensing-communication systems for 6G.

Real-Time State Estimation in Smart Grids over 5G Networks: Experimental Validation Using Raspberry Pis and Typhoon HIL

Biswajit Kumar Dash, Luis Herrera, Filippo Malandra · arXiv eess.SP · 2026-06-26

★☆☆☆☆ Competent vertical application demo but contributes no wireless innovation—treats 5G as a black-box transport layer for smart grid telemetry.

This paper experimentally validates real-time smart grid state estimation over a commercial 5G network using Raspberry Pi nodes and Typhoon HIL simulation. The testbed streams voltage, current, and phase measurements to a remote concentrator, achieving 6.5× lower latency than LTE cat-M and fault detection in 0.80 seconds. The work focuses on characterizing 5G KPIs (delay, jitter, frame loss) for power grid monitoring rather than advancing wireless PHY or access methods.

Contribution: First experimental validation of real-time power grid state estimation over commercial 5G using real hardware and HIL-simulated grid data, moving beyond simulation-only assessments.

Relevance to your work: tangential

Inverse Design of Compact and Wideband Inverted Doherty Power Amplifiers Using Deep Learning

Han Zhou, Haojie Chang, David Widen, Christian Fager · arXiv eess.SP · 2026-06-25

★☆☆☆☆ Competent RF hardware design with ML-assisted optimization, but addresses power amplifier efficiency—a well-studied problem with incremental improvements rather than breakthrough innovation for 6G air interfaces.

This paper applies CNNs and genetic algorithms to inversely design a compact, wideband inverted Doherty power amplifier with a pixelated combiner structure that integrates load modulation, matching, combining, and phase compensation. A GaN HEMT prototype operating over 1.9-2.5 GHz achieves 51-63% peak drain efficiency and 48-54% efficiency at 6-dB back-off. With digital predistortion, the PA demonstrates ACLR better than -53.2 dBc.

Contribution: Novel use of deep learning (CNN) combined with genetic algorithm for inverse synthesis of pixelated combiner networks in Doherty PAs, enabling compact wideband designs that integrate multiple functions in a single structure.

Relevance to your work: Tangential—this is RF front-end hardware optimization for base stations; not directly related to AI-native air interface design, PHY-layer learning, channel modeling, or other core 6G research areas of interest.

Effects of motion cueing on longitudinal acceleration perception in a driving simulator

Erik Gustaf Lilljebjörn, Sogol Kharrazi, Jan Åslund, Martin Singull · arXiv eess.SY · 2026-06-26

★☆☆☆☆ This is automotive human-factors research with no relevance to wireless communication systems, air interface design, or 6G technologies.

This paper evaluates human perception of longitudinal acceleration in a heavy-truck driving simulator, comparing motion-cueing algorithm (MCA) variants tuned for tip-in/launch maneuvers versus general driving. Using psychometric functions and weighted staircase procedures, they found no significant difference in just-noticeable difference (JND) between MCA variants, with mean JND of 5.4% across participants. Participants subjectively preferred the launch-tuned MCAs despite no measurable perception difference.

Contribution: Establishes quantitative human perception thresholds (JND ~5.4%) for longitudinal acceleration in driving simulators, providing psychophysical validation for motion-cueing algorithm tuning in virtual vehicle prototyping.

Relevance to your work: tangential

Host-Driven Flowlet Balancing with Segment Routing over IPv6

Ryo Nakamura, Hiroki Kano, Tomoko Okuzawa · arXiv cs.NI · 2026-06-26

★☆☆☆☆ This is data center networking optimization work with no relevance to wireless systems, physical layer design, or radio access technologies.

This paper proposes a host-driven flowlet load balancing method for data center networks using Segment Routing over IPv6 (SRv6), where hosts detect flowlets and steer them across multiple paths while switches remain stateless. The approach uses a simple model estimating in-flight bytes per path to distribute flowlets evenly. Linux testbed evaluation shows 15% and 33% tail latency reduction versus random flowlet balancing and ECMP respectively.

Contribution: A practical stateless flowlet balancing implementation that pushes traffic steering intelligence to hosts using SRv6, eliminating per-flow state requirements in switches while achieving better load distribution than ECMP.

Relevance to your work: tangential

6G Physical Layer

Multi-Modal Environment-Aware Beam Management for Massive MIMO: A Geometry-Driven Virtual Base Station Framework

Yijie Bian, Wei Guo, Jie Yang, Shenghui Song, et al. · arXiv eess.SP · 2026-06-25

★★★★■ Strong contribution addressing a critical 6G bottleneck (beam training overhead at high frequencies) with a practical multi-modal sensing approach that combines interpretable geometry-based modeling with deep RL, credibly demonstrated in simulation.

This paper proposes a geometry-driven beam management framework for massive MIMO that constructs an offline virtual base station (VBS) database from 3D LiDAR point clouds and user locations, modeling dominant propagation paths through mirror-symmetry reflections off building facades. The VBS enables coarse channel reconstruction without exhaustive beam training, which is refined online via orthogonal pilots with minimal overhead. A hierarchical dual-agent dueling double deep Q-network (DD3QN-CBS) handles multi-user beam selection and interference management, demonstrating improved training efficiency and performance versus baselines.

Contribution: Novel use of LiDAR-derived geometric channel priors to construct interpretable virtual base stations that drastically reduce beam training overhead while maintaining beam management performance through physics-informed channel coarse estimation and targeted refinement.

Relevance to your work: Directly relevant: multi-modal sensing for beam management is a core 6G PHY challenge, and the framework bridges AI/ML (deep RL for beam selection) with physics-based channel modeling using environmental awareness—aligns with your ISAC, massive MIMO, and AI-native PHY interests.

Integrated Sensing and Communications for Real-time Avatar Control in XR over 5G

Nabeel Nisar Bhat, Javad Sameri, Rreze Halili, Rafael Berkvens, et al. · arXiv eess.SP · 2026-06-22

★★★★■ Solid applied ISAC work showing feasibility of dual-use mmWave for XR pose tracking with standard 5G signals, though limited to proof-of-concept body gestures rather than fundamental ISAC PHY innovation.

This work proposes a multimodal XR interaction system combining 5G mmWave ISAC for coarse body-pose estimation and surface electromyography (sEMG) for fine-grained finger gestures. The ISAC component derives power-per-beam-pair (PPBP) metrics from standard 5G NR beam management procedures, achieving 82.2% accuracy on unseen users for body-level gesture recognition. The sEMG sensors capture forearm muscle activity to complement ISAC with finger-level interaction, forming a dual-scale sensing architecture that avoids controllers and line-of-sight camera limitations.

Contribution: First demonstration of repurposing standard 5G NR beam management signals (PPBP) for real-time body-pose estimation in XR, achieving practical accuracy without dedicated sensing waveforms.

Relevance to your work: Directly relevant as a vertical use case validating ISAC feasibility with existing 5G NR beam management primitives, offering empirical data on sensing performance trade-offs for body-scale targets at mmWave.

ISAC (sensing + communication)

ISAC for Sea-Air Networks: Predictive Beam Tracking under Sea Induced Disturbances

Rui Zhang, Fuwang Dong, Wei Wang, Zhen Du · arXiv eess.SY · 2026-06-25

★★★★ Solid incremental contribution applying ISAC principles to a specific maritime scenario, but limited to conventional EKF-based tracking without novel sensing/communication trade-off theory or AI/ML techniques.

This paper addresses beam misalignment in UAV-to-USV communication caused by sea-induced motion disturbances using an ISAC framework that switches between wide-beam sensing and narrow-beam communication modes. An extended Kalman filter predicts the communication receiver's state on the USV by jointly modeling surge, sway, yaw dynamics and sea clutter effects, while optimizing time allocation between sensing and communication. Simulations show improved tracking accuracy over benchmarks in this maritime scenario.

Contribution: First work to address predictive beam tracking in sea-air networks by jointly modeling USV's three-DOF sea-induced motion and sea clutter within an ISAC framework using optimized wide/narrow beam switching.

Relevance to your work: Directly relevant as an ISAC application demonstrating sensing-communication resource allocation in mobile scenarios, though the Kalman filtering approach is conventional compared to emerging learned beam management or neural state prediction methods you explore.